

# Data and Network Security

(Master Degree in Computer Science and Cybersecurity)

Lecture 5  
Spring 2026



# Outline for today

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- Recap last lecture
- Information leakage from ML models
- Privacy preserving learning issues

# Sensitive property

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DATASET

- **Demographic Information:**
  - Age, gender, ethnicity, income level.
- **Behavioral Patterns:**
  - Shopping habits, browsing history, social interactions.
- **Personal Preferences:**
  - Political affiliations, health conditions, lifestyle choices.

# Sensitive property

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DATASET

- **Demographic Information:**
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  - Political affiliations, health conditions, lifestyle choices.

**Disclosure of such properties can lead to privacy breaches, discrimination, or manipulation of individuals.**

# Information leakage from ML models

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Can I infer some (sensitive) property of the dataset used to train an ML model?



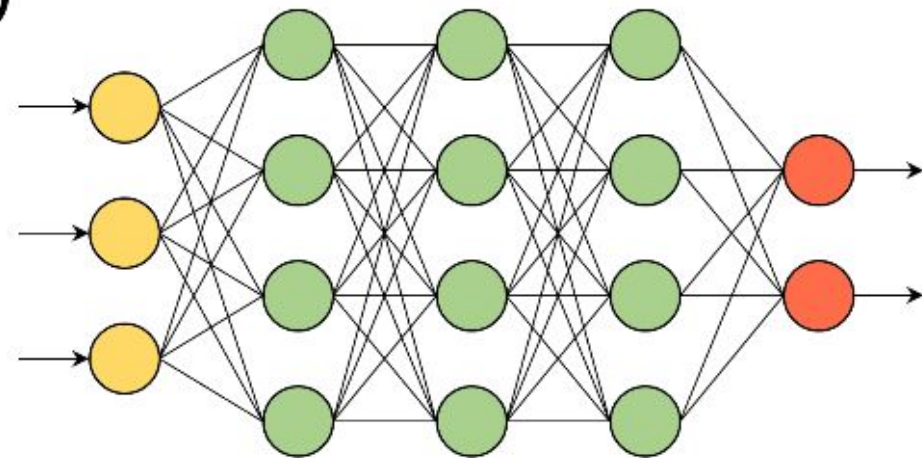
DATASET

# What can ML models tell?



DATASET

(a)



Input layer

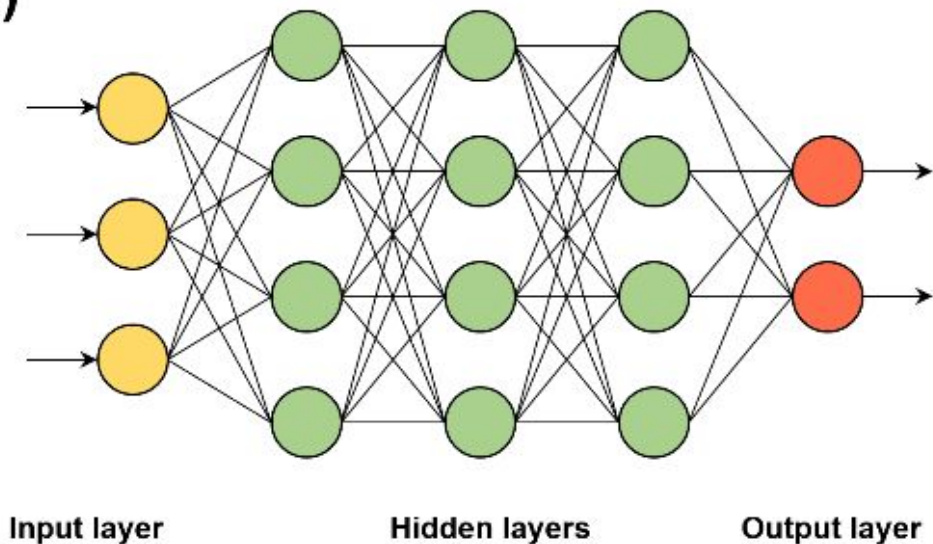
Hidden layers

Output layer

# Does this dataset have this property?

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**(a)**

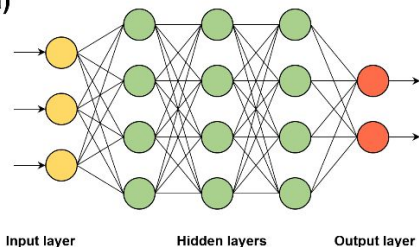


But you have only the model...

# Does this dataset have this property P?

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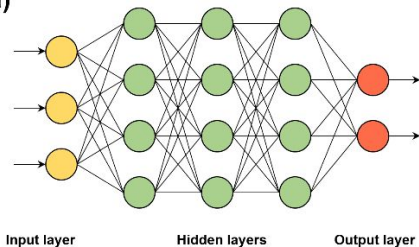
(a)



Construct N different ML models, similar to the target model.

- The dataset of some has property P, and the others dont.

(a)

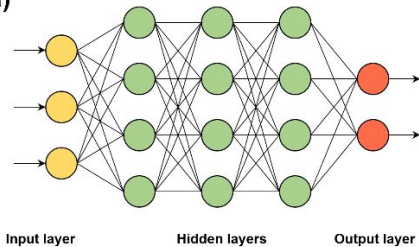




# Does this dataset have this property?

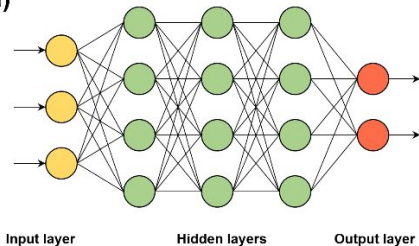
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(a)



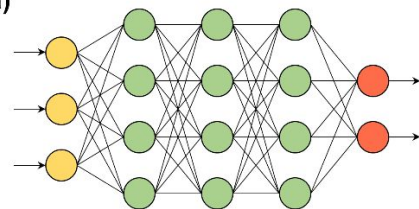
Represent all these ML models  
as a “feature vector”

(a)



# Does this dataset have this property?

(a)



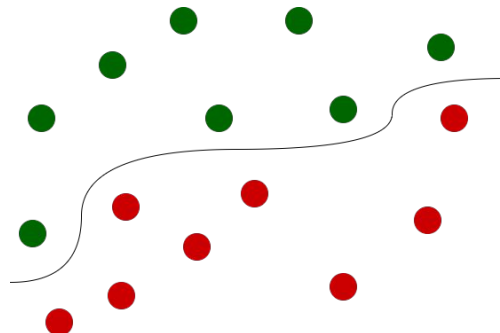
Input layer

Hidden layers

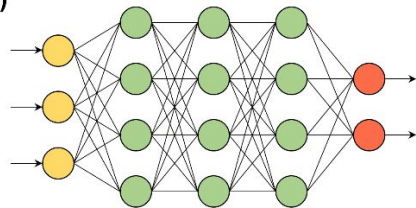
Output layer

$[f_1, f_2, f_3, \dots, f_x]$

Train a binary classifier on these features



(a)



Input layer

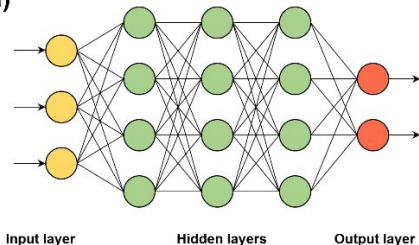
Hidden layers

Output layer

$[f_1, f_2, f_3, \dots, f_x]$

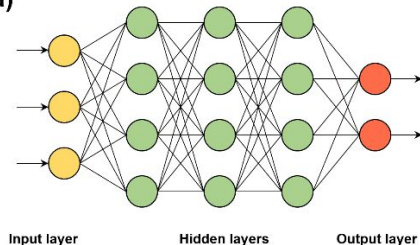
# Does this dataset have this property?

(a)

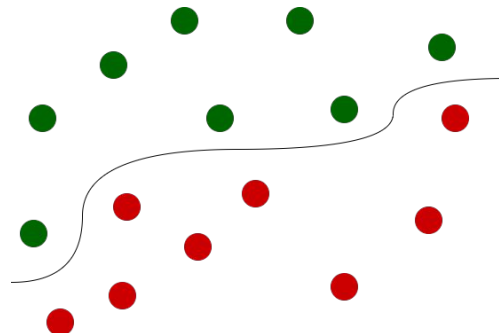


$[f_1, f_2, f_3, \dots, f_x]$

(a)



$[f_1, f_2, f_3, \dots, f_x]$



Represent and test the target classifier.

# Information leakage from ML models

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**Was this datapoint part of this dataset?**

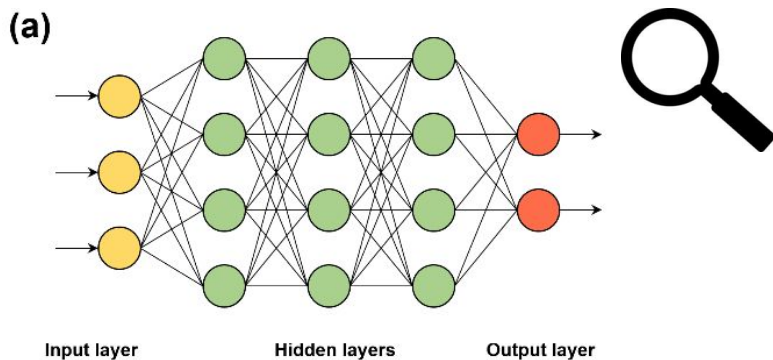


DATASET

# Information leakage from ML models

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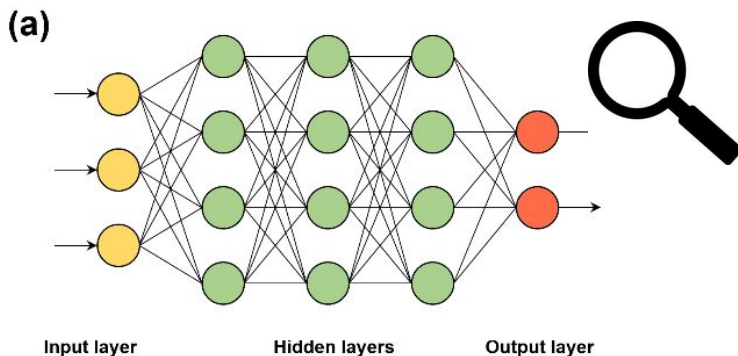
The model's responses provide valuable information that adversaries can leverage to infer whether a particular data point was part of the training dataset.



# Information leakage from ML models

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During the training phase, a machine learning model learns to generalize patterns and relationships from the training dataset to make predictions on unseen data. As a result, the model's behavior may vary depending on whether it has seen a particular data point during training.

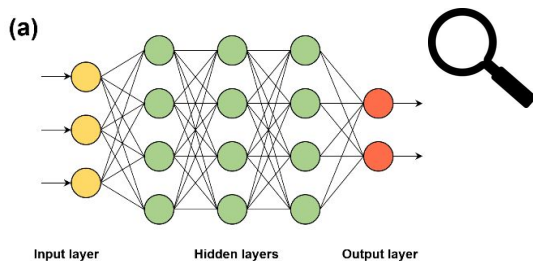


# Information leakage from ML models

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When we query the model with input data, we observe the model's responses:

- predicted labels,
- probabilities, or scores assigned to different classes.

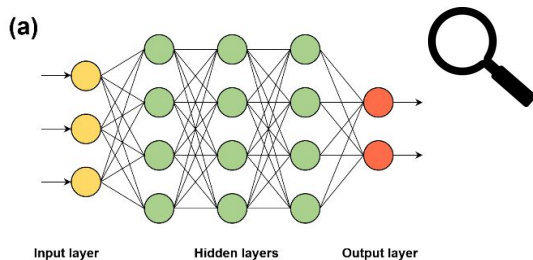


# Information leakage from ML models

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When we query the model with input data, we observe the model's responses:

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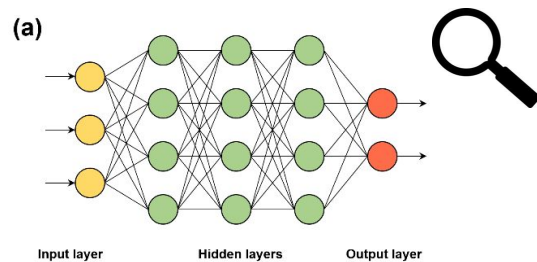
\*distinguishing features in these responses that can indicate whether the input data was likely part of the training dataset



# Distinguishing features?

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One key indicator that a data point was part of the training dataset is overfitting.

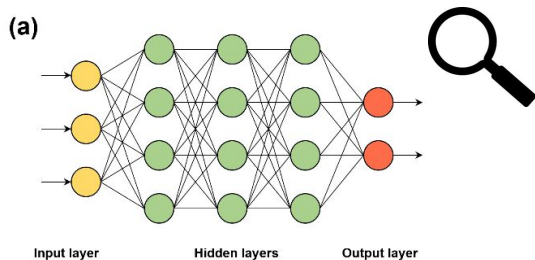


# Distinguishing features?

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One key indicator that a data point was part of the training dataset is overfitting.

Overfitting occurs when a model learns to memorize specific examples from the training data rather than capturing general patterns.



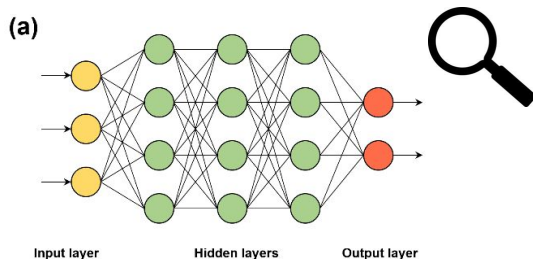
# Distinguishing features?

---

One key indicator that a data point was part of the training dataset is overfitting.

Overfitting occurs when a model learns to memorize specific examples from the training data rather than capturing general patterns.

If a model exhibits overfitting, it may produce responses that are overly **confident** or **precise** for data points seen during training but less accurate for unseen data.

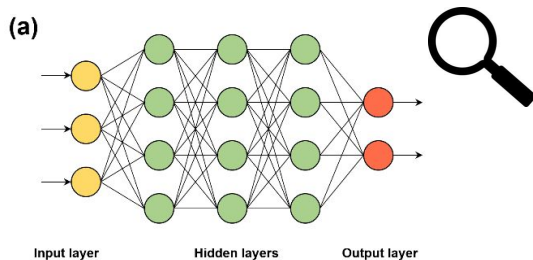


# Distinguishing features?

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Another indicator - confidence discrepancy.

If the model's confidence is significantly higher for certain inputs compared to others, it may suggest that those inputs were present in the training dataset.

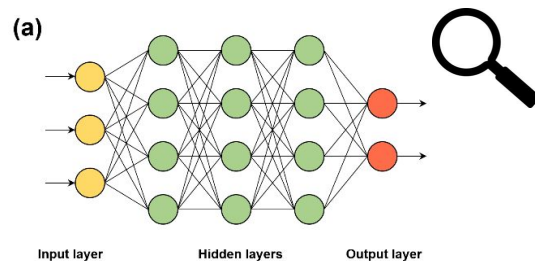


# Distinguishing features?

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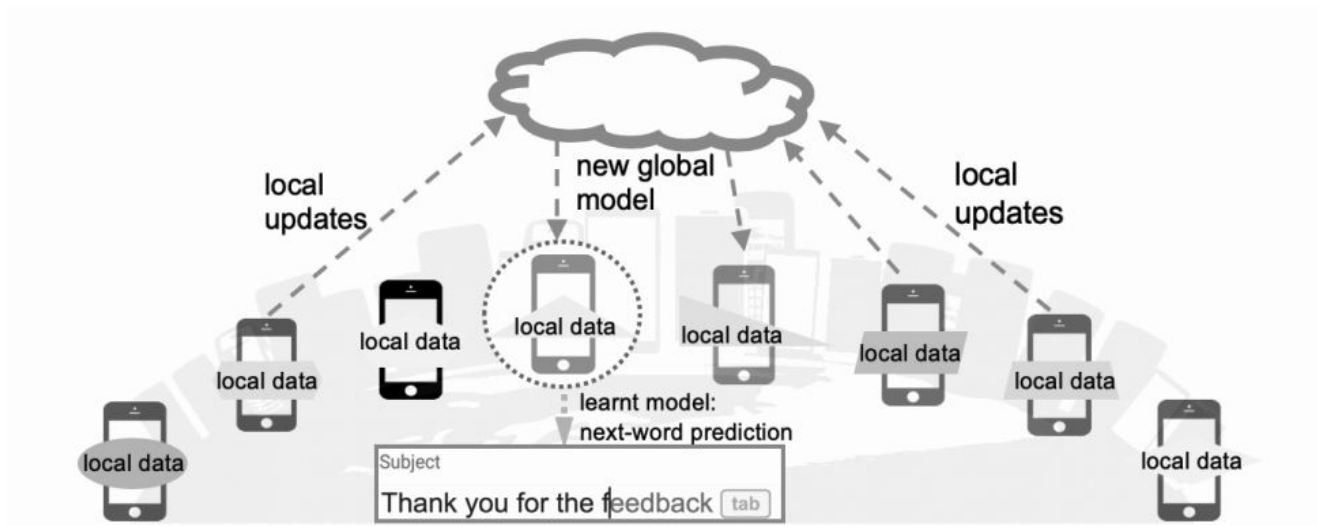
Another indicator - bias in model responses.

If the model consistently predicts certain **classes** or outputs for specific inputs, it may indicate that those inputs were overrepresented in the training dataset.



**Is there any leakage in  
privacy-preserving learning?**

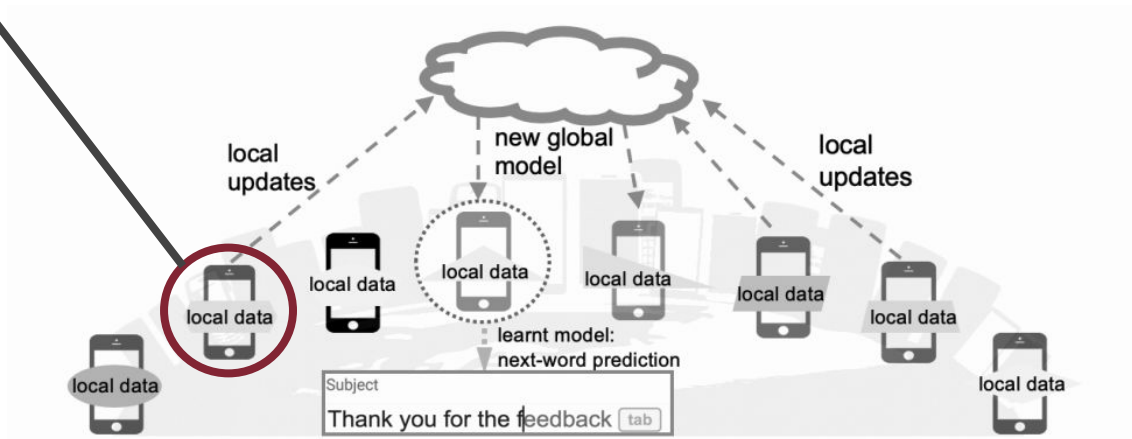
# Collaborative learning



**Federated learning (FL)** (also known as **collaborative learning**) is a machine learning technique that trains an algorithm across multiple decentralized edge devices or servers holding local data samples, without exchanging them.

# How does FL work? (local update)

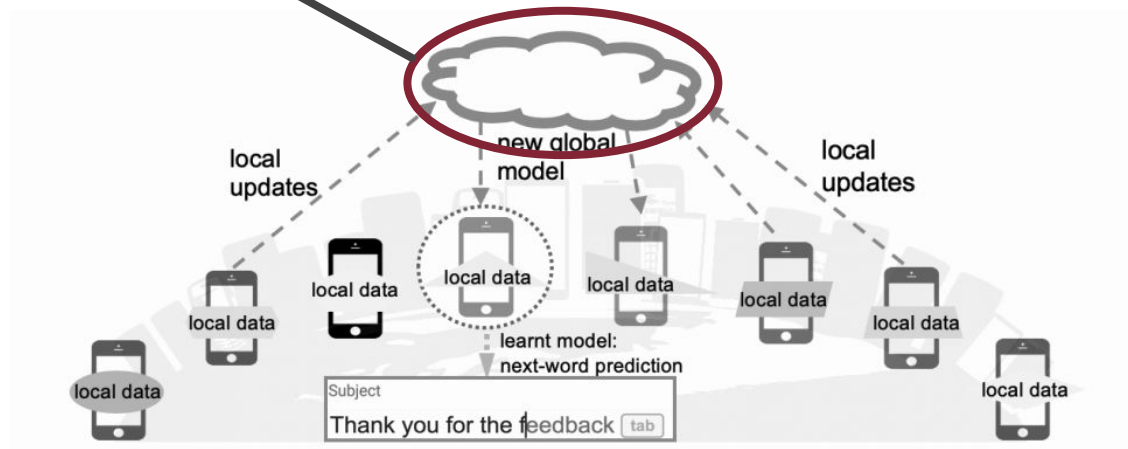
$$\mathbf{W}_{t+1}^k = \mathbf{W}_t + \alpha \nabla \mathbf{W}_t^k$$



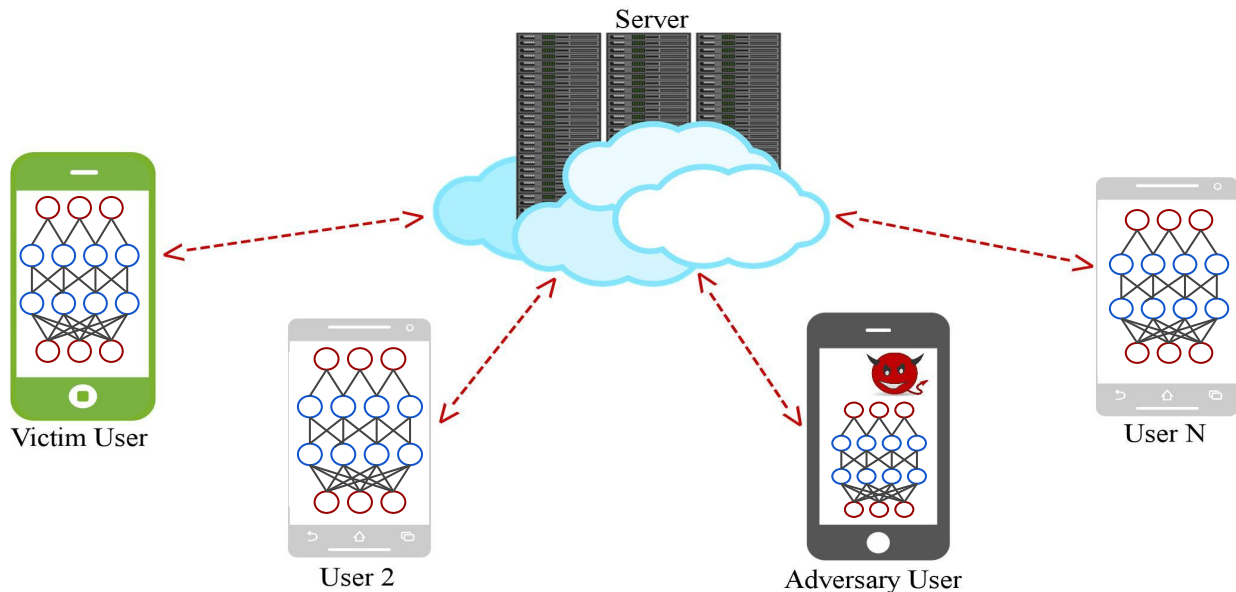


# How does FL work? (global update)

$$\mathbf{W}_{t+1} = \mathbf{W}_t + \frac{\alpha}{n'} \sum_{k=1}^{n'} \nabla \mathbf{W}_t^k$$



# Collaborative Learning Scheme



## Adversary's goal?

Reconstruct private samples from the dataset of the victim indirectly influencing the learning of other participants

**How can we reconstruct samples of other participants training data by looking at some gradients?**

# How should the adversary behave?

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- The adversary should operate as an participant within the privacy-preserving collaborative deep learning protocol.
- The objective of the adversary is to infer meaningful information about a label that he does not own.
- The adversary does not compromise the global parameter server that collects and distributes parameters to the participants.

# What can the adversary use?

— — —

# What can the adversary use?

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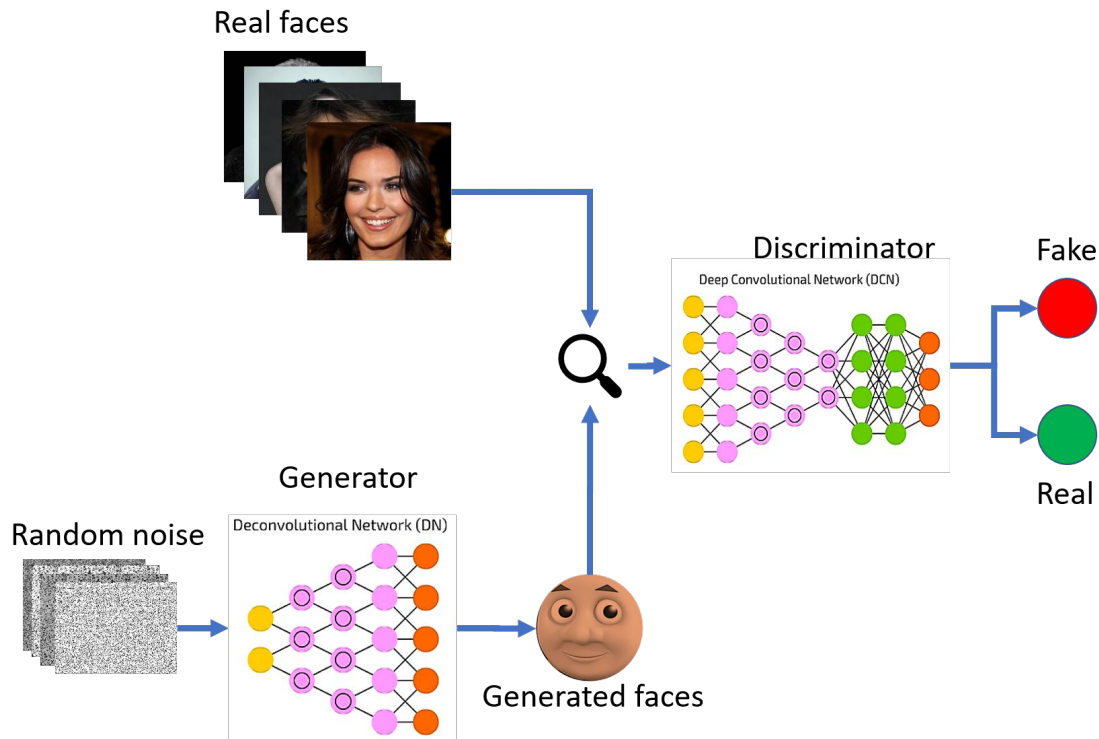
Give me DATA...

# Not this..

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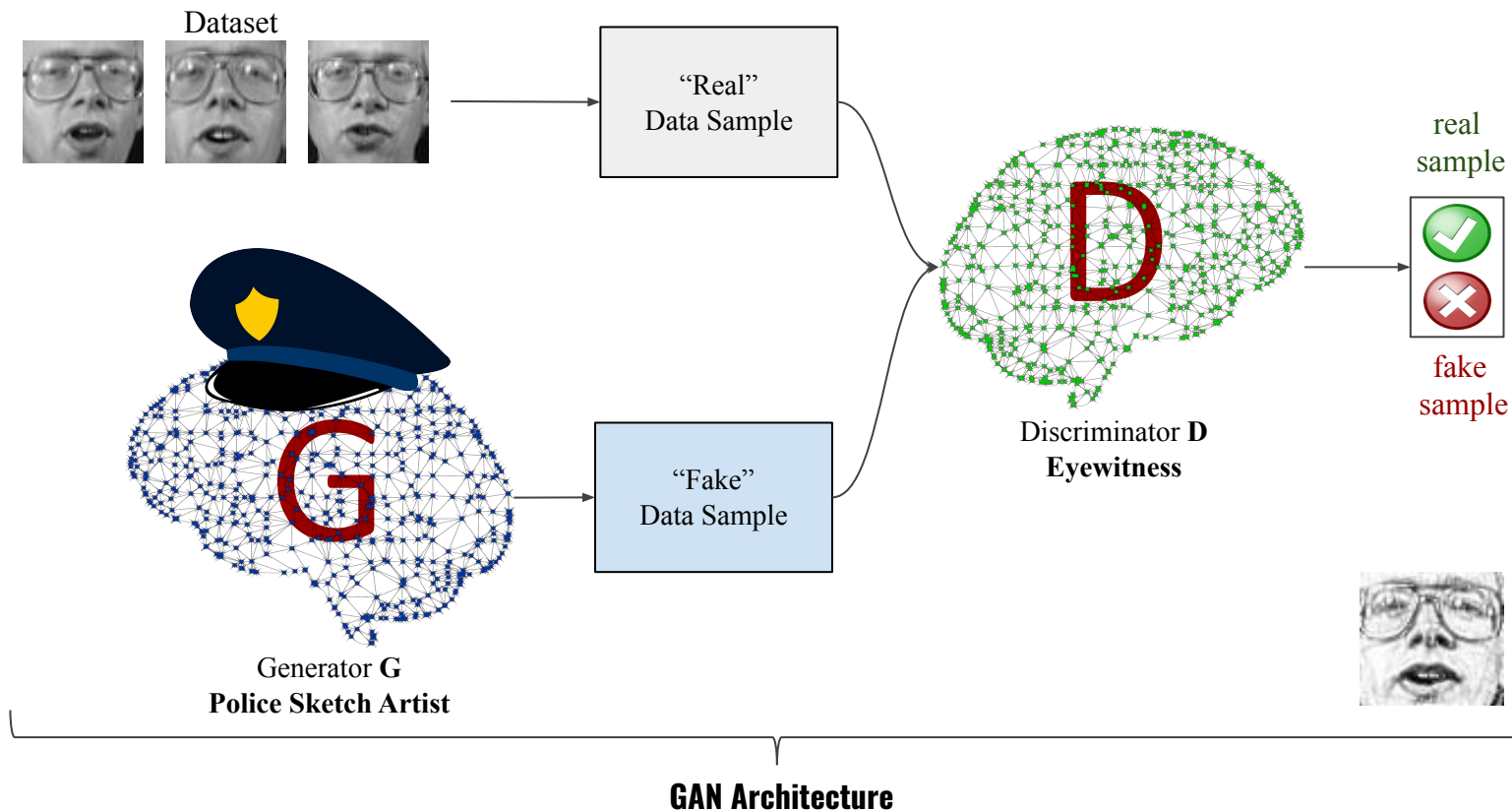


# This...





# Generative Adversarial Network



# Generative Adversarial Networks

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## **Generative**

- We try to learn the underlying the distribution from which our dataset comes from.

# Generative Adversarial Networks

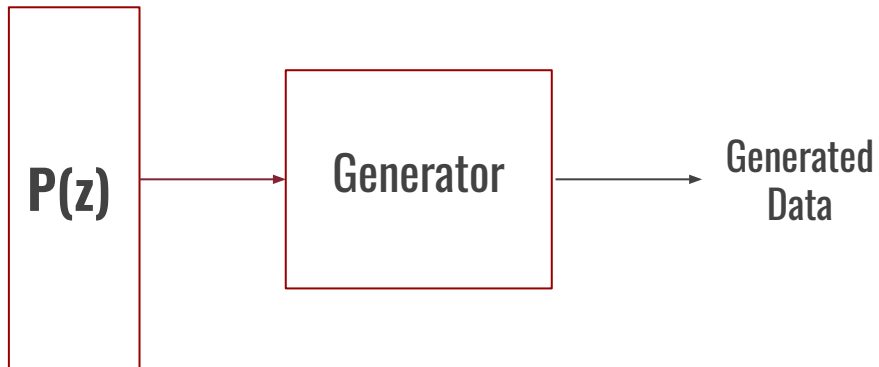
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## Adversarial

- GANS are made up of two competing networks (adversaries) that are trying beat each other.

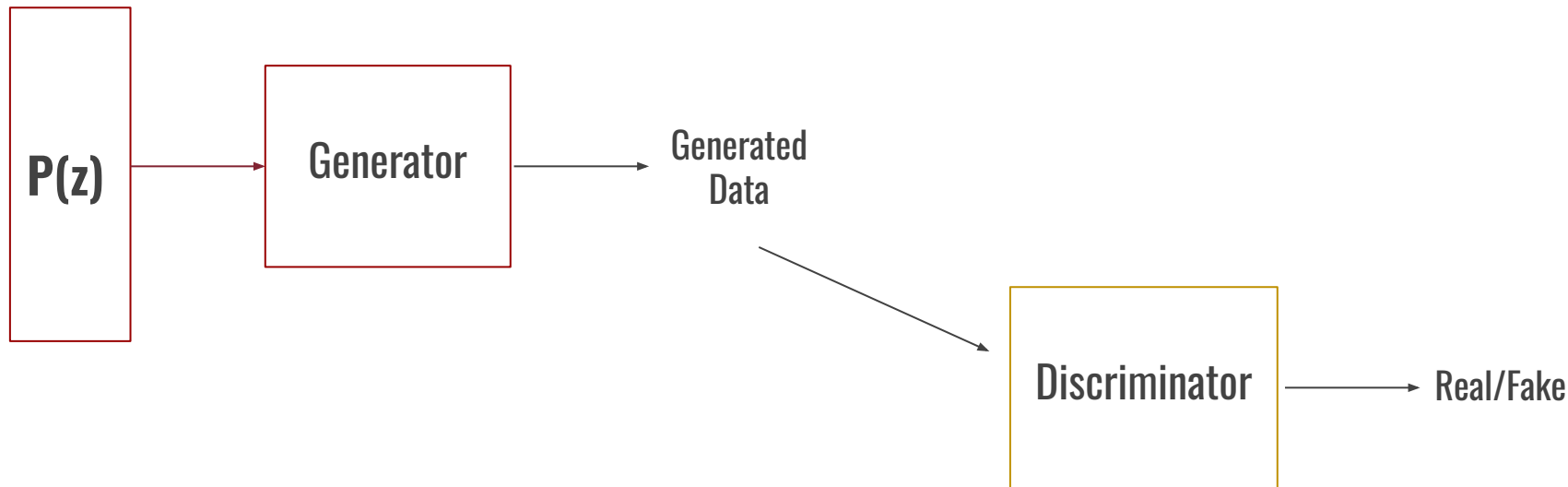
# Generative Adversarial Networks

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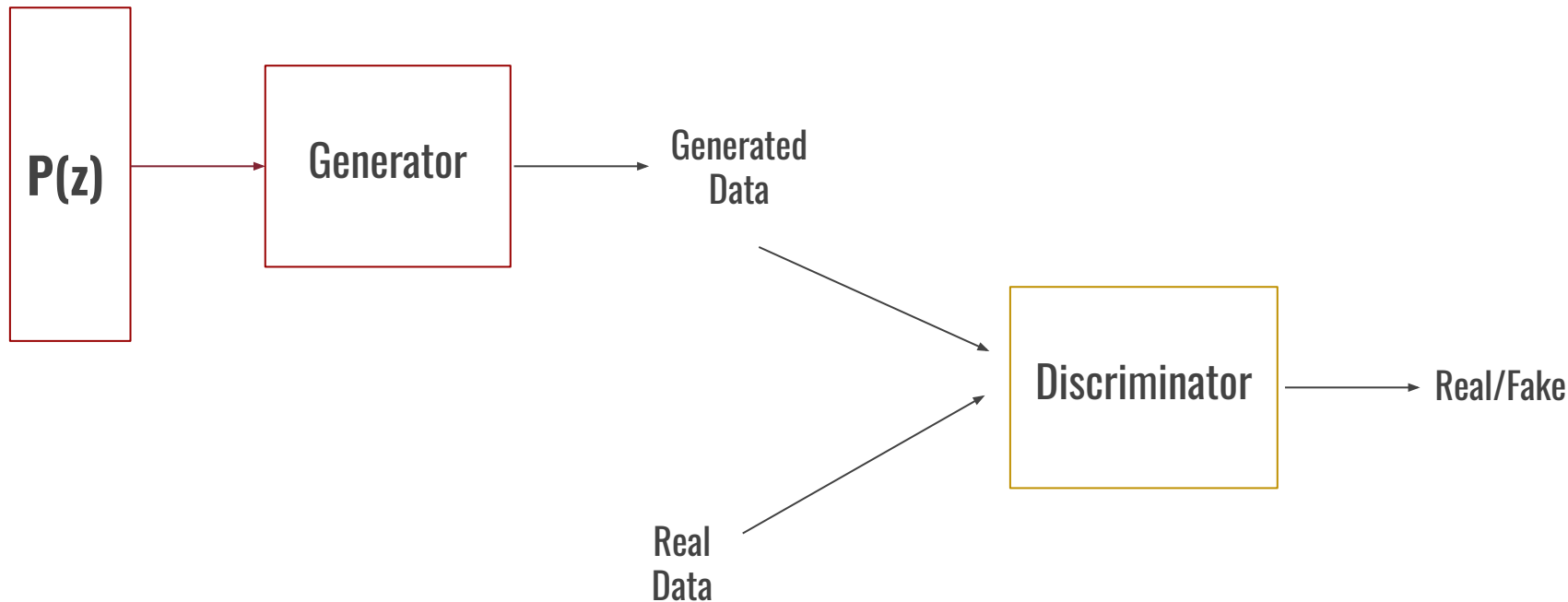
# Generative Adversarial Networks

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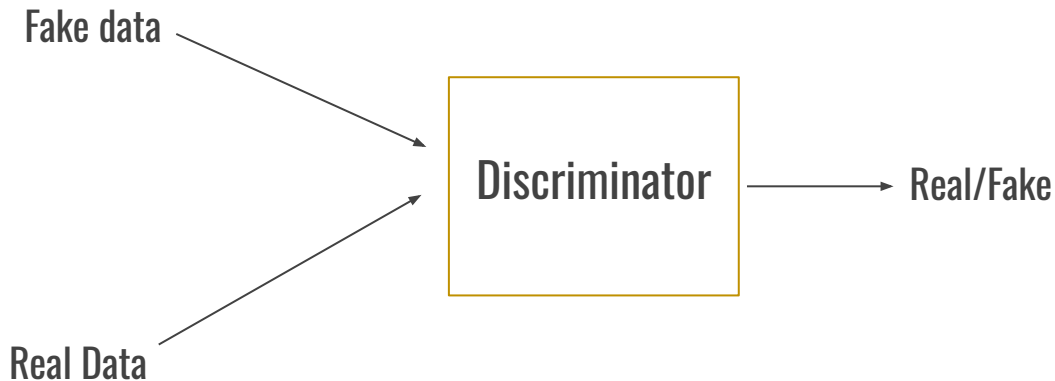
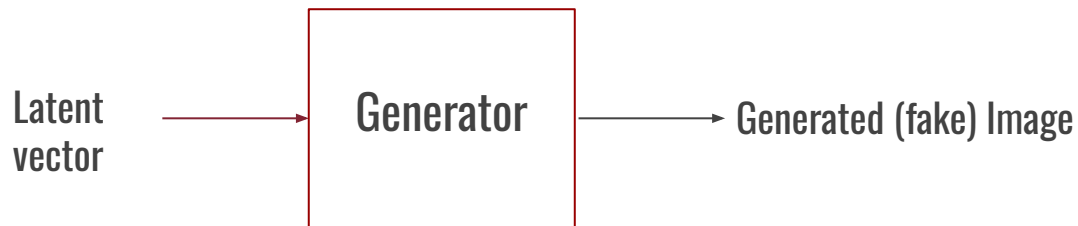
# Generative Adversarial Networks

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# GANs - How are they trained?

At  $t=0$ ,



# GANs - How are they trained?

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Which one should I train first?





# GANs - How are they trained?

---

Which one should I train first?



Discriminator

# GANs - How are they trained?

---

With what training data though?



Discriminator

# GANs - How are they trained?

---

With what training data though?



Discriminator

- The Discriminator is a Binary classifier.
- Needs to discriminate between real/fake

# GANs - How are they trained?

---

With what training data though?



Discriminator

- The Discriminator is a Binary classifier.
- Needs to discriminate between real/fake
- The data for Real class is already given:
  - **THE TRAINING DATA**
- The data for Fake class?
  - Generate from the **Generator**

# GANs - How are they trained?

---

What about the Generator?



# GANs - How are they trained?

---

What about the Generator?



**Learning objective:** Generate images from the Generator such that they are classified incorrectly by the Discriminator.

# GANs - How are they trained?

---



**Discriminator**

Train the Discriminator  
using the current  
ability of the Generator

# GANs - How are they trained?

---

**Discriminator**

Train the Discriminator  
using the current  
ability of the Generator

**Generator**

Train the Generator  
to beat  
the Discriminator

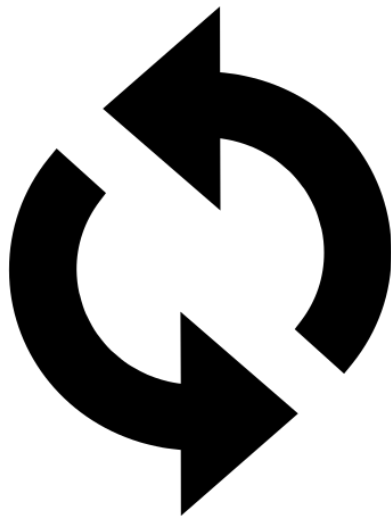


# GANs - How are they trained?

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Discriminator

Train the Discriminator  
using the current  
ability of the Generator

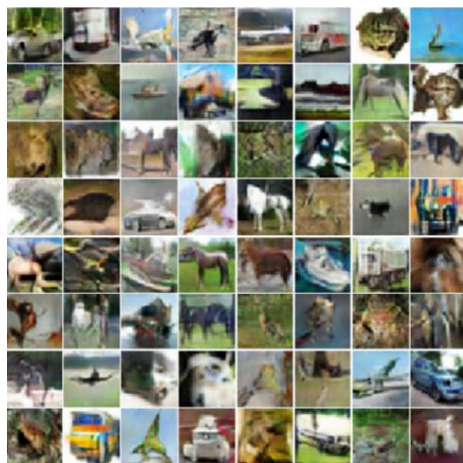


Generator

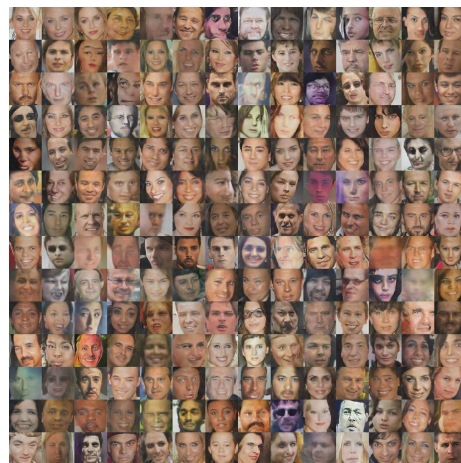
Train the Generator  
to beat  
the Discriminator



MNIST images



CIFAR-10 images



faces



album covers

## GAN results in the literature

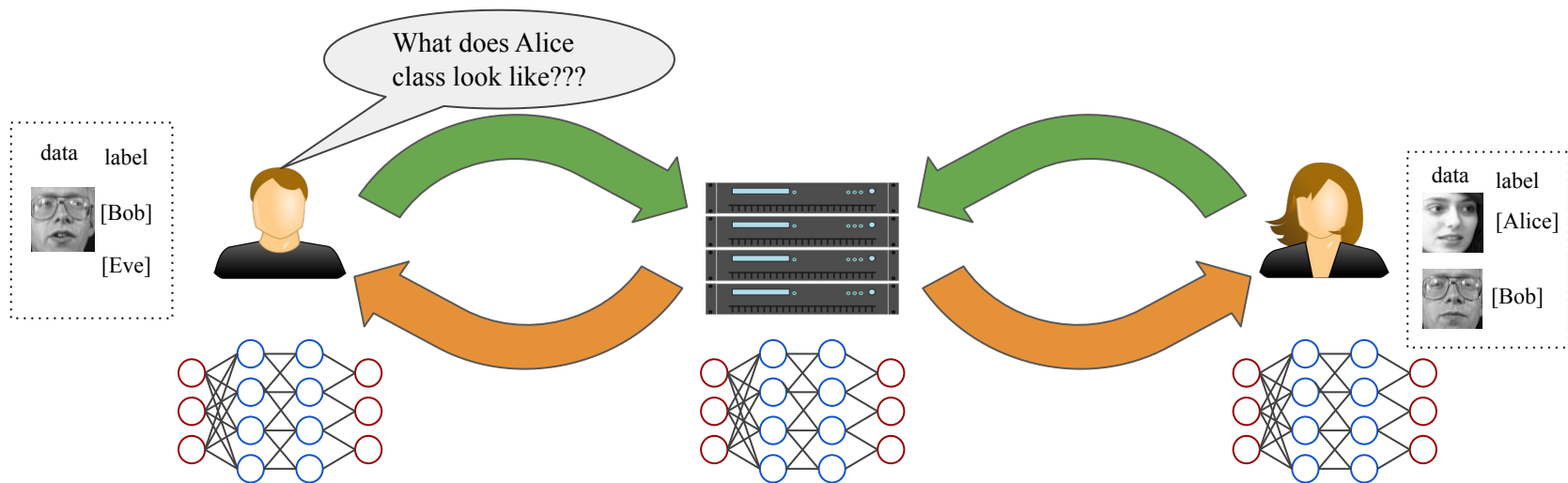
images from:

- <https://blog.openai.com/generative-models/>
- Goodfellow et al. Generative Adversarial Networks
- Radford et al. Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks

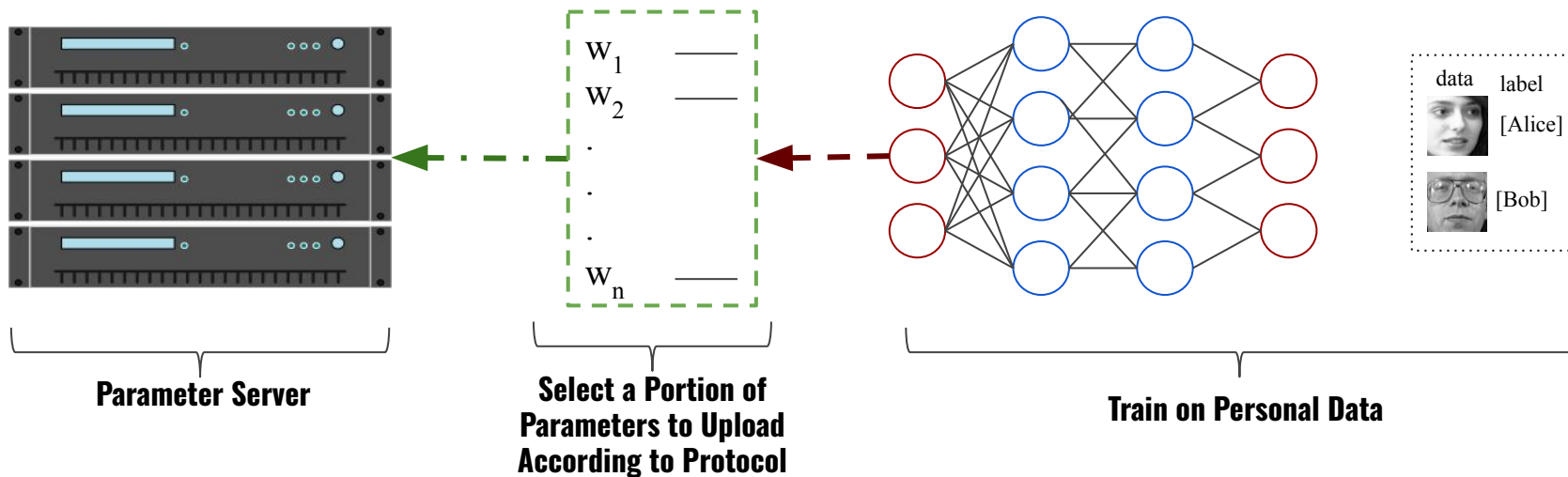
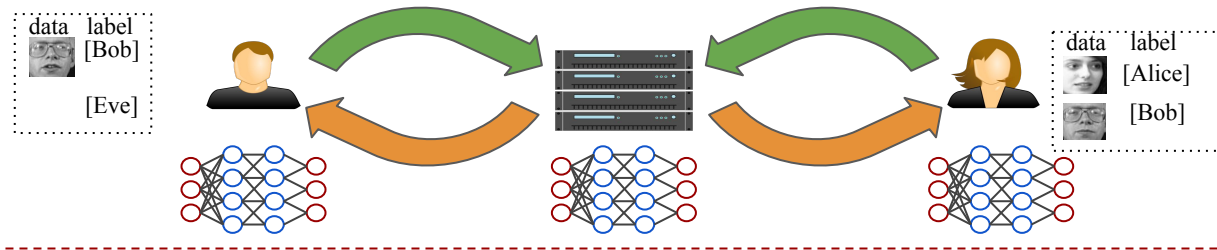


bedrooms

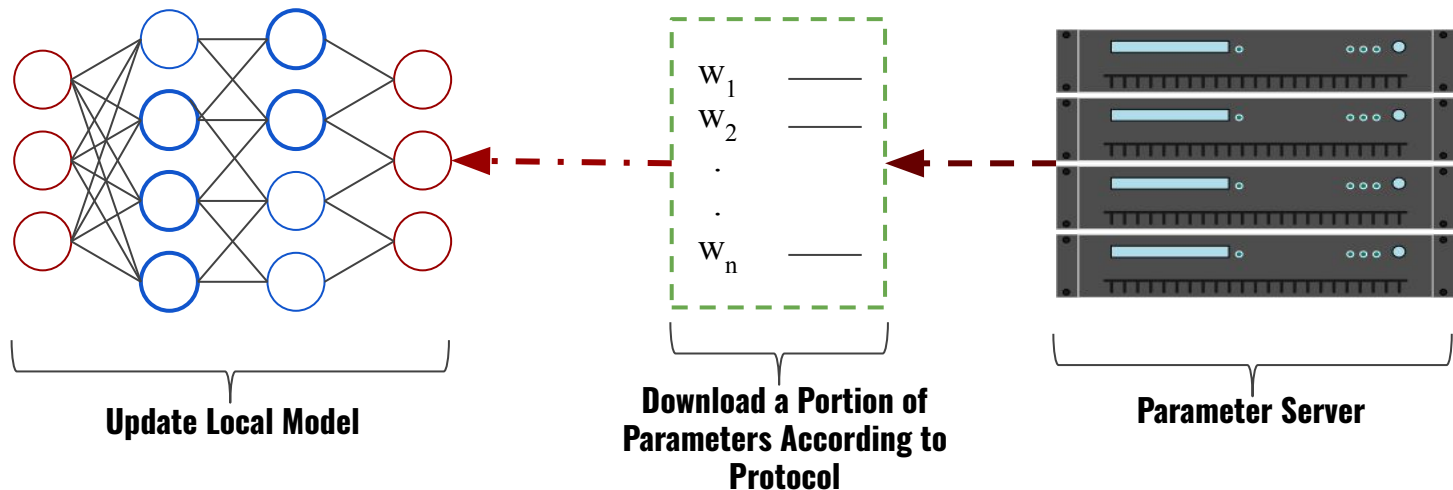
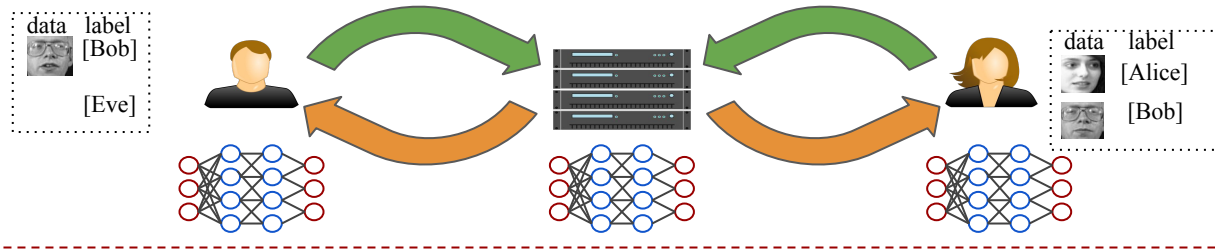
# Violating the privacy...



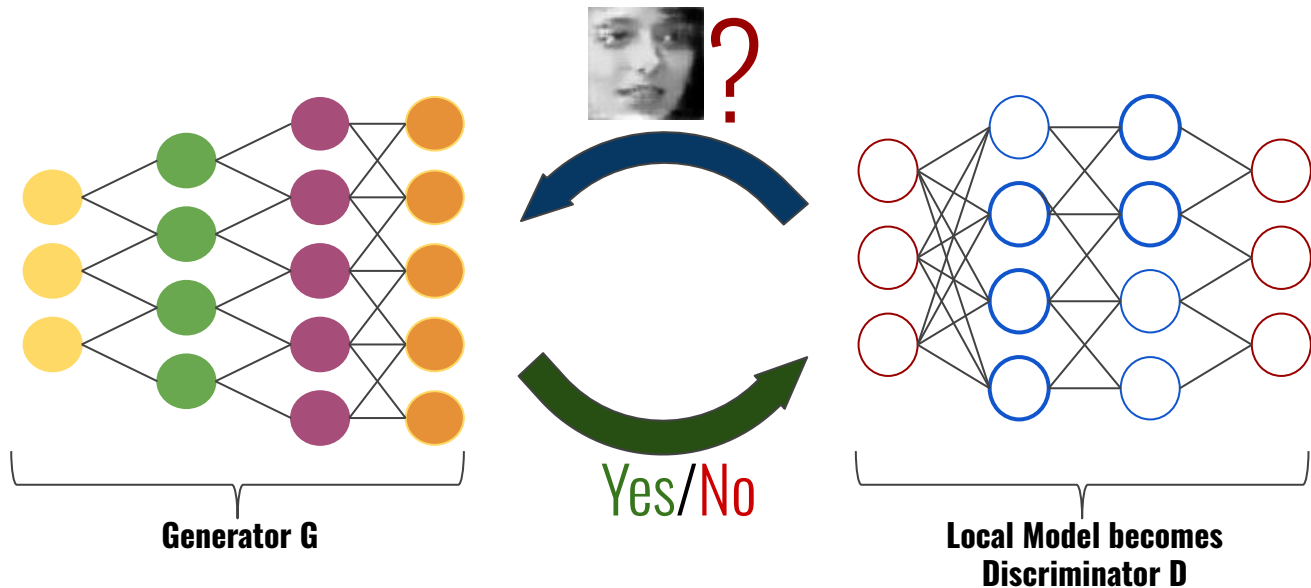
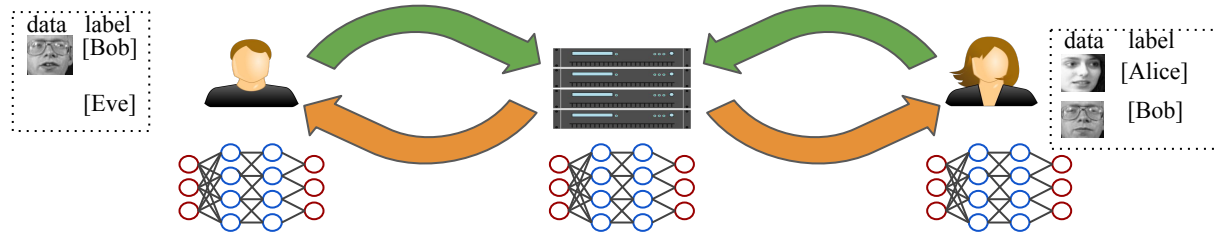
# Victim's Turn



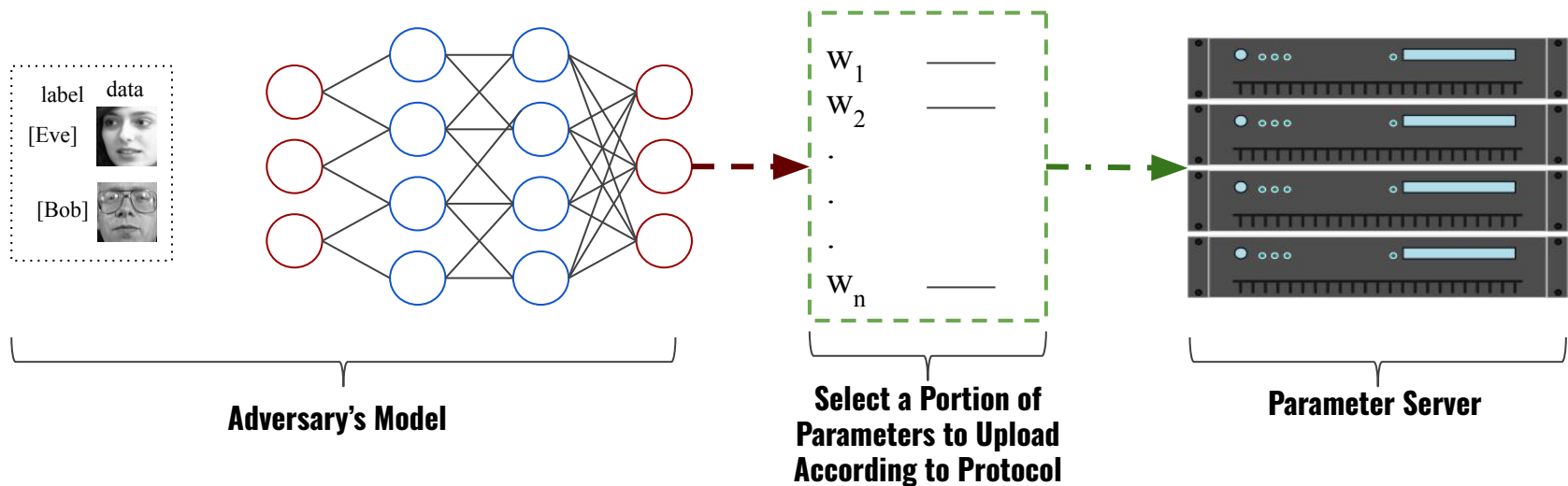
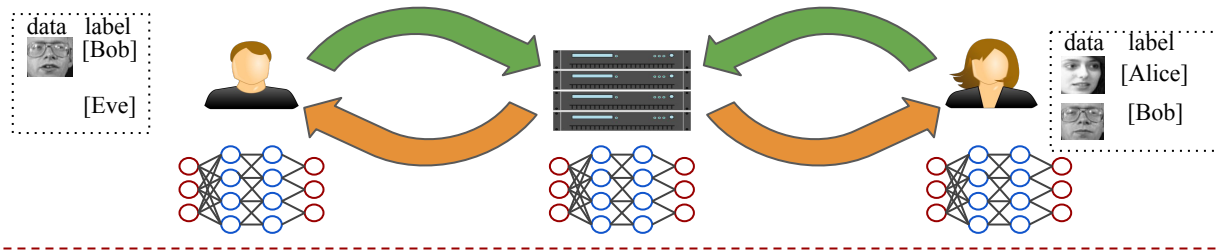
# Adversary's Turn



# Adversary's Turn



# Adversary's Turn



# Experiments without Differential Privacy

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Actual Images



Generated Data



Original vs Generated



# Experiments with Differential Privacy

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Actual Images

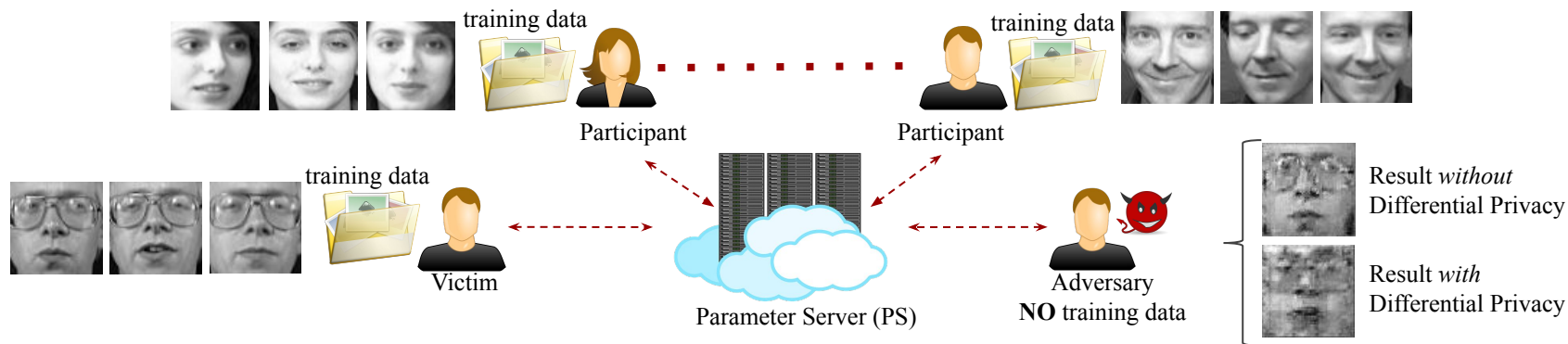


Generated Data



Original vs Generated

# Experiments (Adversary has NO data at all)



# Reading Material

1. Privacy preserving learning: [Link-1](#), [Link-2](#)
2. Generative Adversarial Networks: [Link-1](#)
3. Information Leakage from collaborative learning: [Link-1](#)